Lecture 4. Discrete Random Variables

Expectation, variance, binomial, Poisson, geometric, negative binomial, hypergeometric

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4.1 Random variables and distribution functions

A **random variable** is a real function on S, with **distribution function** $F(x) = P(X \le x)$. It is **discrete** if it takes countably many values $x_1, x_2, \ldots$, with **probability mass function** $p(x) = P(X = x)$, $\sum_i p(x_i) = 1$ and step function $F(a) = \sum_{x_i \le a} p(x_i)$.

Proposition 4.1 (Properties of F)

F is nondecreasing, right-continuous, $F(-\infty) = 0$ and $F(\infty) = 1$. Moreover $P(a < X \le b) = F(b) - F(a)$ and $P(X < b) = \lim_{x \uparrow b} F(x)$.

Proof. Monotonicity: Proposition 2.1(b). As $x_n \downarrow b$, $\{X \le x_n\} \downarrow \{X \le b\}$; as $x_n \uparrow \infty$, $\uparrow S$; as $x_n \uparrow b$, $\uparrow \{X < b\}$. Apply Proposition 2.3. ■

Example 4.1 (Coupon collecting)

Each purchase gives one of N equally likely coupon types, independently; T is the number of purchases to collect all. With A_j: type j missing after n purchases,

$$P(T > n) = P\Big(\bigcup_j A_j\Big) = \sum_{i=1}^{N-1} (-1)^{i+1} \binom{N}{i} \Big(\frac{N-i}{N}\Big)^n,$$

and $P(T = n) = P(T > n-1) - P(T > n)$.

4.2 Expectation

Motivation. The expectation is the long-run average, a one-number summary underlying fair prices, variance and the laws of large numbers.

Definition (Expected value)

$E[X] = \sum_x x p(x)$, provided the sum converges absolutely.

It is the centre of gravity of the pmf; $E[I_A] = P(A)$ for an indicator.

Proposition 4.2

(a) $E[g(X)] = \sum_x g(x)p(x)$.
(b) $E[aX + b] = aE[X] + b$.
(c) On a countable sample space, $E[X] = \sum_{s \in S} X(s)P(\{s\})$, and hence $E[\sum_i X_i] = \sum_i E[X_i]$.

Proof. (a) Group the values x by $y = g(x)$: $\sum_y y P(g(X) = y) = \sum_y \sum_{x:g(x)=y} g(x)p(x) = \sum_x g(x)p(x)$.

(b) is a special case.

(c) Group outcomes s by $X(s)$; linearity of the sum over s gives the second claim. ■

In general $E[g(X)] \ne g(E[X])$.

Example 4.2 (Which question first?)

A contestant answers question 1 or 2 first and continues only if correct; question i is correct with probability p_i independently and worth V_i. Order 1–2 gives $p_1(1-p_2)V_1+p_1p_2(V_1+V_2) = p_1V_1 + p_1p_2V_2$, order 2–1 gives $p_2V_2 + p_1p_2V_1$; so 1 first iff

$$\frac{p_1V_1}{1-p_1} > \frac{p_2V_2}{1-p_2}.$$

Example 4.3 (The friendship paradox)

Pick a random friendship and one of its ends; that person has D^* friends, with $P(D^* = k) = k\,P(D = k)/E[D]$, D the degree of a uniform person (a person with k friends is k times as likely reached). Hence

$$E[D^*] = \frac{E[D^2]}{E[D]} \geq E[D],$$

with equality only for equal degrees: on average your friends have more friends than you.

4.3 Variance

Definition (Variance)

With $\mu = E[X]$, $\mathrm{Var}(X) = E[(X-\mu)^2] = E[X^2] - \mu^2$. The standard deviation is $\sqrt{\mathrm{Var}(X)}$.

(Expand the square.) So $E[X^2] \geq (E[X])^2$, and $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$. Fair die: $E[X] = 7/2$, $E[X^2] = 91/6$, $\mathrm{Var}(X) = 35/12$.

4.4 Bernoulli and binomial random variables

$X \sim \mathrm{Bin}(n,p)$ counts successes in n independent trials with success probability p:

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n.$$

(each of the $\binom{n}{k}$ sequences with k successes has probability $p^k(1-p)^{n-k}$). $\mathrm{Bin}(1,p)$ is Bernoulli.

Proposition 4.3

If $X \sim \mathrm{Bin}(n,p)$ then $E[X^k] = np\,E[(Y+1)^{k-1}]$ with $Y \sim \mathrm{Bin}(n-1,p)$. Hence $E[X] = np$ and $\mathrm{Var}(X) = np(1-p)$.

Proof. Using $i\binom{n}{i} = n\binom{n-1}{i-1}$,

$$E[X^k] = \sum_{i=1}^{n} i^{k-1}\, n \binom{n-1}{i-1} p^i (1-p)^{n-i} = np \sum_{j=0}^{n-1} (j+1)^{k-1} \binom{n-1}{j} p^j (1-p)^{n-1-j}.$$

$k = 1$ gives np; $k = 2$ gives $E[X^2] = np\{(n-1)p+1\}$, so $\mathrm{Var}(X) = np(1-p)$. ∎

Since $p(k)/p(k-1) = \frac{(n-k+1)p}{k(1-p)}$, the pmf increases while $k < (n+1)p$, then decreases: mode $\lfloor (n+1)p \rfloor$. The ratio also gives a stable cdf recursion.

Example 4.4 (Chuck-a-luck)

Bet 1 on a number; three dice are rolled; win i if it appears $i \geq 1$ times, else lose 1. Appearances are $\text{Bin}(3, 1/6)$, so the expected gain is

$$-\tfrac{125}{216} + 1 \cdot \tfrac{75}{216} + 2 \cdot \tfrac{15}{216} + 3 \cdot \tfrac{1}{216} = -\tfrac{17}{216} \approx -0.079.$$

Example 4.5 (k-out-of-n systems)

n components work independently with probability p; the system works if at least half do. $P_5 = \binom{5}{3}p^3q^2 + 5p^4q + p^5$, $P_3 = 3p^2q + p^3$, and $P_5 - P_3 = 3p^2(p-1)^2(2p-1)$: five components are better iff $p > \frac{1}{2}$.

4.5 The Poisson random variable

Motivation. Rare-event counts are approximately Poisson, the limit of binomials with large n and small p.

$X \sim \text{Poisson}(\lambda)$ if $p(k) = e^{-\lambda}\lambda^k/k!$, $k \geq 0$; $E[X] = \text{Var}(X) = \lambda$ (shift trick: $E[X] = \lambda \sum_j e^{-\lambda}\lambda^j/j!$, $E[X^2] = \lambda E[X+1]$).

Proposition 4.4 (Poisson approximation)

If $n \to \infty$ and $p \to 0$ with $np = \lambda$ fixed, then $\binom{n}{k}p^k(1-p)^{n-k} \to e^{-\lambda}\lambda^k/k!$.

Proof. $\binom{n}{k}p^k(1-p)^{n-k} = \dfrac{n(n-1)\cdots(n-k+1)}{n^k} \cdot \dfrac{\lambda^k}{k!} \cdot \dfrac{(1-\lambda/n)^n}{(1-\lambda/n)^k}$, and the three factors tend to 1, $\lambda^k/k!$ and $e^{-\lambda}$. ∎

So counts of rare events among many trials are approximately Poisson, also for *weakly dependent* trials with small unequal probabilities, $\lambda = \sum p_i$ (the Poisson paradigm).

Example 4.6 (Poisson paradigm)

(a) Matching: the weakly dependent events "person i gets own hat" have probability $1/N$, so matches are approximately Poisson(1) (cf. Example 2.4).
(b) Birthdays: the $\binom{n}{2}$ weakly dependent pair events have probability $1/365$, so $P(\text{no shared birthday}) \approx \exp\{-n(n-1)/730\}$; $n = 23$ gives 0.500 (exact 0.493).

Example 4.7 (Poisson process)

Events occur in time with (i) independent counts on disjoint intervals, (ii) one event in an interval of length h with probability $\lambda h + o(h)$, two or more with $o(h)$. Splitting $[0, t]$ into n pieces, $N(t) \approx \text{Bin}(n, \lambda t/n)$, and as $n \to \infty$

$$P(N(t) = k) = e^{-\lambda t}\frac{(\lambda t)^k}{k!}.$$

4.6 Other discrete distributions

Geometric. Trials up to the first success: $p(k) = (1-p)^{k-1}p$, $k \geq 1$, $P(X > k) = (1-p)^k$. With $q = 1-p$, conditioning on the first trial, $E[X] = p + q(1 + E[X])$ gives $E[X] = 1/p$, and $E[X^2] = p + q\,E[(1+X)^2]$ gives $E[X^2] = (2-p)/p^2$, $\mathrm{Var}(X) = q/p^2$.

Negative binomial. Trials needed for r successes ($r-1$ successes in the first $k-1$ trials, then a success):

$$p(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k \geq r, \qquad E[X] = \frac{r}{p}, \quad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}.$$

(A sum of r independent geometrics; see Lecture 7.)

Example 4.8 (Banach match problem)

A smoker has two boxes of N matches and picks one at random each time. When a box is first found empty, what is the probability the other has k matches? With "left box" a success, the left box is found empty with k on the right when success $N+1$ occurs at trial $2N-k+1$; doubling for symmetry, $2\binom{2N-k}{N}(\frac{1}{2})^{2N-k+1} = \binom{2N-k}{N}(\frac{1}{2})^{2N-k}$.

Hypergeometric. White balls X among n drawn without replacement from N, m white:

$$P(X=i) = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}}, \qquad E[X] = \frac{nm}{N}, \qquad \mathrm{Var}(X) = n\frac{m}{N}\Big(1-\frac{m}{N}\Big)\frac{N-n}{N-1}.$$

With $p = m/N$: the binomial mean, and variance with finite-population correction $(N-n)/(N-1)$. For $N \gg n$, $X \approx \mathrm{Bin}(n, m/N)$.

Example 4.9 (Capture–recapture)

Tag m animals, later catch n, of which i are tagged. $P_i(N) = \binom{m}{i}\binom{N-m}{n-i}/\binom{N}{n}$ satisfies

$$\frac{P_i(N)}{P_i(N-1)} = \frac{(N-m)(N-n)}{N(N-m-n+i)} \geq 1 \iff N \leq \frac{mn}{i},$$

so the maximum likelihood estimate is $\lfloor mn/i \rfloor$ (setting $i/n = m/N$).

Zeta (Zipf). $p(k) = C/k^{\alpha+1}$, $k \geq 1$, $C = 1/\zeta(\alpha+1)$; heavy-tailed frequencies such as word counts.

4.7 Expected value of sums

By Proposition 4.2(c), $E[X_1 + \cdots + X_n] = \sum E[X_i]$ for *any* random variables. Writing counts as sums of indicators is the main trick.

Example 4.10

(a) The sum of n fair dice has mean $7n/2$.
(b) If trial i succeeds with probability p_i, the expected number of successes is $\sum p_i$.

(c) Hypergeometric: with I_j indicating that white ball j is drawn, $E[X] = m \cdot n/N$ (Example 2.1(b)).

Example 4.11 (Variance of the number of successes)

$X = \sum_i I_i$ gives $E[X^2] = \sum_i p_i + \sum_{i \neq j} P(I_i = I_j = 1)$. Independent trials: $E[X^2] = np + n(n-1)p^2$, $\mathrm{Var}(X) = np(1-p)$. Hypergeometric: $P(I_i = I_j = 1) = \frac{m(m-1)}{N(N-1)}$, giving the variance above.

4.8 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): re-indexing binomial sums, conditioning on a count, and links among Poisson, gamma, binomial and negative binomial.

Exercise 4.10 (A reciprocal moment of the binomial)

Let $X \sim \mathrm{Bin}(n, p)$ with $0 < p \leq 1$. Show that

$$E\Big[\frac{1}{X+1}\Big] = \frac{1-(1-p)^{n+1}}{(n+1)p}.$$

Solution.

Step 1. For $0 \leq i \leq n$,

$$\frac{1}{i+1}\binom{n}{i} = \frac{n!}{(i+1)!\,(n-i)!} = \frac{1}{n+1}\binom{n+1}{i+1}.$$

Step 2. With $q = 1-p$ and Proposition 4.2(a),

$$\begin{aligned} E\Big[\frac{1}{X+1}\Big] &= \sum_{i=0}^{n} \frac{1}{i+1}\binom{n}{i} p^i q^{n-i} = \frac{1}{(n+1)p} \sum_{i=0}^{n} \binom{n+1}{i+1} p^{i+1} q^{n-i} \\ &= \frac{1}{(n+1)p} \sum_{j=1}^{n+1} \binom{n+1}{j} p^j q^{n+1-j} = \frac{(p+q)^{n+1} - q^{n+1}}{(n+1)p}, \end{aligned}$$

by the binomial theorem (Theorem 1.3) without the $j = 0$ term; $p + q = 1$. ∎

Exercise 4.25 (Thinning a Poisson count)

The number N of events in a fixed period is Poisson(λ), and each event is counted with probability p, independently of the other events and of N. Show that the number M of counted events is Poisson(λp). If $\lambda = 10$ uranium deposits are expected in an area and each is discovered with probability $1/50$, find the probability that at most one deposit is discovered.

Solution.

Step 1. Given $N = n$, $M \sim \mathrm{Bin}(n, p)$: $P(M = k \mid N = n) = \binom{n}{k} p^k q^{n-k}$ $(k \leq n)$, $q = 1-p$.

Step 2. By total probability over the countable partition $\{N = n\}$ (Theorem 3.2),

$$P(M = k) = \sum_{n=k}^{\infty} e^{-\lambda} \frac{\lambda^n}{n!} \cdot \frac{n!}{k!\,(n-k)!} p^k q^{n-k}$$

$$= e^{-\lambda}\frac{(\lambda p)^k}{k!}\sum_{j=0}^{\infty}\frac{(\lambda q)^j}{j!} = e^{-\lambda}\frac{(\lambda p)^k}{k!}e^{\lambda q} = e^{-\lambda p}\frac{(\lambda p)^k}{k!}.$$

Step 3. (Intuition: $N \approx \mathrm{Bin}(n, \lambda/n)$, so $M \approx \mathrm{Bin}(n, \lambda p/n) \to \mathrm{Poisson}(\lambda p)$ by Proposition 4.4.)

Step 4. Here $\lambda p = 0.2$, so $P(M \le 1) = e^{-0.2}(1 + 0.2) \approx 0.982$. ■

Exercise 4.26 (Poisson tails as gamma integrals)

Prove that for $\lambda > 0$ and $n = 0, 1, 2, \ldots,$

$$\sum_{i=0}^{n} e^{-\lambda}\frac{\lambda^i}{i!} = \frac{1}{n!}\int_{\lambda}^{\infty} e^{-x}x^n\,dx.$$

Solution.

Step 1. Let $I_n = \frac{1}{n!}\int_{\lambda}^{\infty} e^{-x}x^n\,dx$; $I_0 = e^{-\lambda}$.

Step 2. For $n \ge 1$ integrate by parts ($u = x^n/n!$, $dv = e^{-x}dx$; $x^n e^{-x} \to 0$):

$$I_n = \Big[-e^{-x}\frac{x^n}{n!}\Big]_{\lambda}^{\infty} + \int_{\lambda}^{\infty} e^{-x}\frac{x^{n-1}}{(n-1)!}\,dx = e^{-\lambda}\frac{\lambda^n}{n!} + I_{n-1}.$$

Step 3. Induct on n. Equivalently, $P(X \le n) = P(Y > \lambda)$ for $X \sim \mathrm{Poisson}(\lambda)$, $Y \sim \mathrm{Gamma}(n + 1, 1)$: in a rate-one Poisson process, at most n events by time λ iff event $n + 1$ comes after λ. ■

Exercise 4.28 (Negative binomial versus binomial)

Let X be negative binomial with parameters (r, p) and $Y \sim \mathrm{Bin}(n, p)$. Show that $P(X > n) = P(Y < r)$.

Solution.

Step 1. In one sequence of independent trials, let X be the trial of the r-th success and Y the successes among the first n; these have the stated laws (Section 4.6), so show $\{X > n\} = \{Y < r\}$.

Step 2. $Y \ge r$ iff the first n trials contain r successes iff $X \le n$. Take complements. ■